

SC Honors Geometry - Nappier
Homework 10.3 – Using Chords

Name: _____

Chords and arcs.

1. A central angle measures 81° . Find its minor arc.
2. In one circle, chords $RS \cong TU$ and $m\widehat{RS} = 117^\circ$. Find $m\widehat{TU}$.
3. A diameter is perpendicular to a 30-unit chord. How long is each piece of the chord?

Right-triangle chord problems (draw the radius!).

4. Radius 17, distance from center to chord 8. Find the chord.
5. Radius 10, chord 16. Find the distance from the center to the chord.
6. A 24-unit chord is 9 units from the center. Find the radius.

Equidistant chords.

7. Chords $5x - 9$ and $3x + 7$ are equidistant from the center. Find x and the chord length.
8. Two congruent chords sit in a circle. What do you know about their distances to the center?
9. A diameter perpendicular to a chord splits its arc into $(2y + 10)^\circ$ and $(4y - 20)^\circ$. Find y and each arc piece.
10. An archaeologist finds a broken circular plate. A half-chord of 20 sits 21 units from where the center must be. Find the plate's radius.